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1.1 研究背景

Agent

Agent Agent

“ ” “ ”

1.2 研究问题

Agent

1.3 研究内容与贡献

AEIC Low AEIC

Verify Imitate Prompt Token

Follower Trace

Agent

P2P

1.4 论文结构

AEIC

ReDial

2.1 多智能体审议与答案聚合

Judge Self-consistency

Agent Agent

2.2 结构化推理与工具使用

Chain-of-Thought Tree-of-Thoughts ReAct

Agent

2.3 机制设计与领导者追随者关系

m=(g,c,P,B)

Follower

2.4 影视推荐与会话式检索

Agent MovieLens ReDial

3.1 参与者与状态

Leader Follower Agent Leader Follower i x m h_i π_i(a_i|x,m,h_i) Agent

3.2 行动空间与资源路径

A={Low, AEIC, Verify, Imitate} Low AEIC Verify AEIC Imitate Prompt Token

3.3 时序

t t-1 Agent Trace

3.4 集体决策

Conclusion Top-K

AEIC

4.1 AEIC 报告

Agent i r_i=(A_i,E_i,I_i,C_i) A E I C

4.2 可观察信号

complete_aeic consistency field_A field_E field_I field_C novel_evidence verified_evidence useful_correction unsupported_flip Token verified_evidence unsupported_flip

4.3 信用、成本与效用

credit_i=Σ_k w_k s_k(r_i,h_i,y) s_k w_k Agent utility_i=credit_i-cost_i reputation_t=decay×reputation_(t-1)+utility_t

4.4 可证伪预测

Follower

5.1 数据与条件

StrategyQA MMLU-Hard TruthfulQA 150 Agent
No Mechanism Outcome Only Static AEIC

5.2 指标与统计

AEIC Verify Token
McNemar Holm 3 5
Agent

5.3 复现与数据完整性

5,850 270 240 6,360
SHA-256 JSONL

6.1 主实验

No Mechanism 0.46% 68.00% Static AEIC 56.93% 56.47 82.22% 14.22 78 14 Holm McNemar p=1.23×10^-11 Agent-round 145.5 Token

Outcome Only 3.08% 2.62 67.33% -0.67 Follower

6.2 分域结果

StrategyQA 73.33% 77.33% MMLU-Hard 71.33% 91.33% TruthfulQA 59.33% 78.00% StrategyQA MMLU-Hard TruthfulQA

6.3 机制检查与稳健性

Static AEIC

Seed Panel

6.4 报告接口、等计算量与后端边界

Token AEIC AEIC AEIC Token GLM

7.1 系统需求

Agent Trace

7.2 总体架构

MovieLens Preference Agent
Diversity Agent Critic Agent AEIC
Top-K

7.3 混合召回

18,430 ReDial
30 0.85

7.4 硬约束与共识

Agent
Agent

7.5 可审计 Trace 与失败降级

Trace request_id Agent AEIC Token
DeepSeek Agent

8.1 工程验收

30 150 Top-5 Trace Agent 100%

8.2 ReDial 基准构建与防泄漏

ReDial 8,004 1,342 MovieLens 7,348 1,203

ID

8.3 全量召回结果

1,203 Hit@5 Recall@5 NDCG@5 0.75% 0.26% 0.24% 10.47% 5.83% 4.88% 0.85 10.81% 5.95% 5.01% 10.06 Bootstrap 95% [8.40,11.89] 123 2 Exact McNemar p=3.70×10^-34

8.4 Agent 重排结果

100 Top-10 Agent Agent Hit@5 6% 7% 7% Recall@5 4.17% 5.17% 5.17% NDCG@5 2.66% 3.11% 2.96% Agent McNemar p=1.0 100%

Agent 5,842 Token 7.44 Agent 16,935 Token 12.68 P95 28.69 51.2%

8.5 失败归因

130 Top-5 626 52.04% 21 300 Top-50 24.94% 93 Top-50 Top-10 33 Top-10 Top-5 Agent

9.1 机制实验与系统实验的关系

Agent

Trace ReDial AEIC

9.2 有效范围

Prompt Agent Prompt

DeepSeek

9.3 局限

AEIC Token

9.4 后续工作

Follower Leader

AEIC

Static AEIC AEIC

Top-K ReDial

Agent Agent Trace

`references.bib` StrategyQA MMLU

TruthfulQA DeepSeek MovieLens ReDial